

Lecture 3: Linear Models for Regression

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- 1 Linear Regression
 - Gradient Descent (GD) variants for Linear Regression
 - Normal Equations and Least Squares
 - Probabilistic Interpretation of Linear Regression
- 2 Under-fitting, Polynomial Regression, and Over-fitting
 - Polynomial Features and Fitting Behavior
- 3 Regularization: Ridge Regression, and Lasso Regression
 - Ridge Regression
 - Lasso Regression
 - Another View of Regularization: Geometric Interpretation
 - Ridge Regression as MAP Estimator

Reading materials

- Chapter 1, Stanford CS 229 Lecture Notes,
https://cs229.stanford.edu/notes2022fall/main_notes.pdf
- Chapter 3.1 & 3.2, Bishop, Pattern Recognition and Machine Learning
- Read about over-fitting in the paper by Pedro Domingos (Sections 3 and 5 of “A few useful things to know about machine learning”)

Reference

- EPFL, CS-433 Machine Learning, https://github.com/epfml/ML_course

Session 1: How do we solve linear regression?

LIVE QA — think, vote, explain

How can we solve linear regression—and when should we not form an inverse?

Three questions about linear regression

- 1 **How do we solve it?** MSE, GD/SGD, normal equations, and projection.
- 2 **What does the loss mean?** Noise, likelihood, polynomial features, and fitting behavior.
- 3 **How do we control complexity?** Ridge, lasso, and a probabilistic view.

Takeaway

Today's goal: choose a regression method and explain the choice.

LIVE QA routine: Think first, discuss your reasoning, then compare with the worked answer. **Next lecture:** generalization gap, model selection, and bias–variance.

Linear regression: predictions first, then residuals

Read the shapes: N examples, D features, one weight per feature.

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times D}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} \in \mathbb{R}^D.$$

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Predict: $f_{\mathbf{w}}(\mathbf{X}) = \hat{\mathbf{y}} = \mathbf{X}\mathbf{w}$, so $\hat{y}_n = \mathbf{x}_n^\top \mathbf{w}$.

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\mathbf{w}, \quad e_n = y_n - \mathbf{x}_n^\top \mathbf{w}.$$

Interpret: a positive residual means the prediction is too small.

From residuals to MSE, then to a gradient

Choose weights that make the residuals small. The factor $1/2$ simplifies derivatives but does not change the minimizer.

$$\mathcal{L}(\mathbf{w}) = \frac{1}{2N} \sum_{n=1}^N \underbrace{(y_n - \mathbf{x}_n^\top \mathbf{w})^2}_{e_n^2} = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2.$$

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Takeaway

Batch GD follows the negative gradient: $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + \frac{\gamma}{N} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w}^{(t)})$, with learning rate $\gamma > 0$.

A tiny regression step: follow every quantity

Question: for one example, will gradient descent move the prediction up or down?

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad y = 2$$

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$$e = y - \hat{y} = 3 > 0.$$

The prediction is too small.

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This is one-example loss, so there is no extra $1/N$.

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Takeaway

$\mathbf{w}^+ = \mathbf{w} - \gamma \nabla \ell(\mathbf{w})$ increases both weights and moves $\hat{y}$ upward for a sufficiently small $\gamma > 0$.

Stochastic Gradient Descent (SGD): how much data per step?

Write $\mathcal{L} = \frac{1}{N} \sum_n \mathcal{L}_n$, where $\mathcal{L}_n(\mathbf{w}) = \frac{1}{2}(y_n - \mathbf{x}_n^\top \mathbf{w})^2$. Every method uses $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \gamma \mathbf{g}^{(t)}$.

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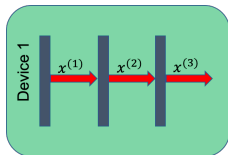
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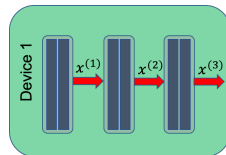
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SGD: cheap, noisy steps.

All gradients in a mini-batch use the *same* current weights.



Mini-batches: average several gradients.

Why can one example guide the whole objective?

LIVE QA — 15 s: vote yes/no, then justify

If we sample n uniformly from $\{1, \dots, N\}$, is its gradient the full gradient on every step?

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Let the random index be $I \sim \text{Uniform}\{1, \dots, N\}$. One update uses the random vector $\nabla \mathcal{L}_I(\mathbf{w}^{(t)})$. **It is not the full gradient on each draw, but it is unbiased:**

$$\mathbb{E}_I[\nabla \mathcal{L}_I(\mathbf{w}^{(t)}) \mid \mathbf{w}^{(t)}] = \sum_{i=1}^N \frac{1}{N} \nabla \mathcal{L}_i(\mathbf{w}^{(t)}) = \nabla \mathcal{L}(\mathbf{w}^{(t)}).$$

A uniformly sampled mini-batch average is unbiased as well.

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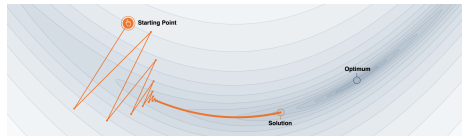
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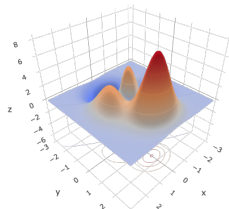
SGD trades a noisy direction for less work per update. Mini-batches average noise; the learning rate still controls stability.

Motivation: can we find the optimum without repeated steps?

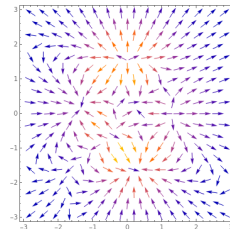
GD/SGD approach a minimizer iteratively. Least squares also has an analytical characterization.



Iterative updates



Generic function



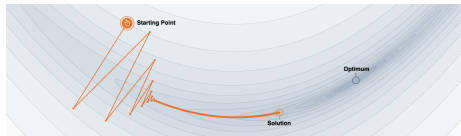
Its gradient field

LIVE QA — 20 s solo, then compare with a neighbor

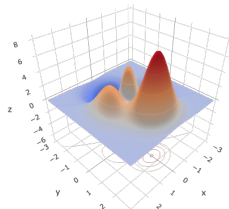
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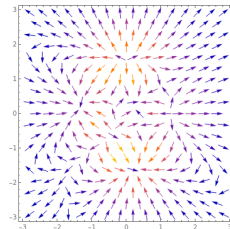
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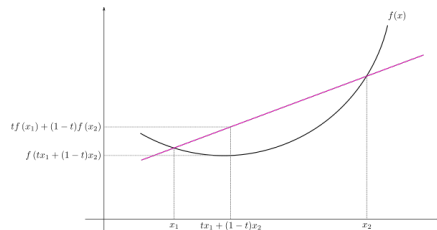
For the MSE, first justify that a zero gradient gives a global minimum; then solve $\nabla \mathcal{L}(\mathbf{w}^*) = 0$.

Steps to form normal equations: why convexity matters

Convexity: the line segment between two points on the graph never lies below the graph.

$$h(\lambda \mathbf{u} + (1 - \lambda) \mathbf{v}) \leq \lambda h(\mathbf{u}) + (1 - \lambda) h(\mathbf{v}),$$

for $0 \leq \lambda \leq 1$.

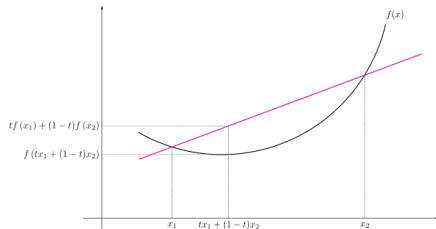


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Takeaway

For a differentiable convex loss, a zero gradient is sufficient for a global minimum. Convexity alone does not guarantee uniqueness.

Next: verify this property for our MSE loss.

Derivation (step 1): the MSE is convex in the weights

Differentiate the gradient once more:

$$\nabla \mathcal{L}(\mathbf{w}) = \frac{1}{N}(\mathbf{X}^\top \mathbf{X} \mathbf{w} - \mathbf{X}^\top \mathbf{y}), \quad \nabla^2 \mathcal{L}(\mathbf{w}) = \frac{1}{N} \mathbf{X}^\top \mathbf{X}.$$

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Test any direction $\mathbf{v} \in \mathbb{R}^D$:

$$\mathbf{v}^\top \nabla^2 \mathcal{L}(\mathbf{w}) \mathbf{v} = \frac{1}{N} (\mathbf{X} \mathbf{v})^\top (\mathbf{X} \mathbf{v}) = \frac{1}{N} \|\mathbf{X} \mathbf{v}\|_2^2 \geq 0.$$

The Hessian is positive semidefinite, so the MSE is convex.

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If $\mathbf{X}$ has full column rank, this quantity is positive for $\mathbf{v} \neq \mathbf{0}$: the loss is strictly convex and the minimizer is unique.

Derivation (step 2): set the gradient to zero

Keep the residual convention $\mathbf{e} = \mathbf{y} - \mathbf{X}\mathbf{w}$.

$$\nabla \mathcal{L}(\mathbf{w}^*) = -\frac{1}{N} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w}^*) = \mathbf{0}$$

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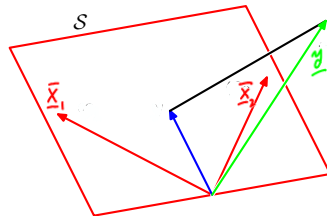
These are the [normal equations](#). Convexity makes every solution an MSE minimizer, even when the coefficients are not unique.

Geometric interpretation of Normal Equations

Which predictions are possible?

$$\mathcal{S} = \text{col}(\mathbf{X}) = \{\mathbf{X}\mathbf{w} : \mathbf{w} \in \mathbb{R}^D\} \subseteq \mathbb{R}^N.$$

This is the span of the *columns* of $\mathbf{X}$; each column is a vector across all N examples.



Bishop: projection in data space

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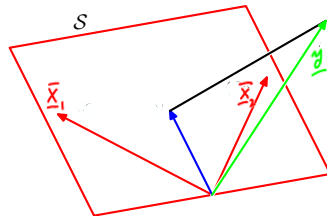
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$$\mathbf{X}^\top \mathbf{e}^* = \mathbf{0} \implies \mathbf{e}^* \perp \text{col}(\mathbf{X}).$$

The residual is orthogonal to every feature column.



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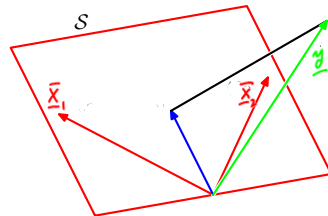
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Takeaway

$\hat{\mathbf{y}}^* = \mathbf{X}\mathbf{w}^*$ is the orthogonal projection of $\mathbf{y}$ onto $\text{col}(\mathbf{X})$: the closest attainable prediction.

Why does an orthogonal residual minimize squared error?

CHECKPOINT QA — predict yes/no before the algebra

Starting at $\hat{\mathbf{y}}^*$, change the weights by $\Delta \mathbf{w}$. Can the new prediction $\hat{\mathbf{y}}^* + \mathbf{X}\Delta \mathbf{w}$ be closer to $\mathbf{y}$?

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$$\begin{aligned}\|\mathbf{y} - (\hat{\mathbf{y}}^* + \mathbf{X}\Delta \mathbf{w})\|_2^2 &= \|\mathbf{e}^* - \mathbf{X}\Delta \mathbf{w}\|_2^2 \\ &= \|\mathbf{e}^*\|_2^2 - 2(\mathbf{e}^*)^\top \mathbf{X}\Delta \mathbf{w} + \|\mathbf{X}\Delta \mathbf{w}\|_2^2.\end{aligned}$$

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Normal equations give $\mathbf{X}^\top \mathbf{e}^* = \mathbf{0}$, hence $(\mathbf{e}^*)^\top \mathbf{X}\Delta \mathbf{w} = 0$.

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CHECKPOINT QA — predict yes/no before the algebra

Starting at $\hat{\mathbf{y}}^*$, change the weights by $\Delta \mathbf{w}$. Can the new prediction $\hat{\mathbf{y}}^* + \mathbf{X}\Delta \mathbf{w}$ be closer to $\mathbf{y}$?

$$\begin{aligned}\|\mathbf{y} - (\hat{\mathbf{y}}^* + \mathbf{X}\Delta \mathbf{w})\|_2^2 &= \|\mathbf{e}^* - \mathbf{X}\Delta \mathbf{w}\|_2^2 \\ &= \|\mathbf{e}^*\|_2^2 - 2(\mathbf{e}^*)^\top \mathbf{X}\Delta \mathbf{w} + \|\mathbf{X}\Delta \mathbf{w}\|_2^2.\end{aligned}$$

Normal equations give $\mathbf{X}^\top \mathbf{e}^* = \mathbf{0}$, hence $(\mathbf{e}^*)^\top \mathbf{X}\Delta \mathbf{w} = 0$.

$$\|\mathbf{y} - (\hat{\mathbf{y}}^* + \mathbf{X}\Delta \mathbf{w})\|_2^2 = \|\mathbf{e}^*\|_2^2 + \|\mathbf{X}\Delta \mathbf{w}\|_2^2 \geq \|\mathbf{e}^*\|_2^2.$$

Answer: no. A different fitted vector increases the squared error. A weight change with $\mathbf{X}\Delta \mathbf{w} = \mathbf{0}$ leaves the fit unchanged.

Least Squares: a closed form when columns are independent

Assumption: $\text{rank}(\mathbf{X}) = D$ (full column rank). Then the Gram matrix $\mathbf{X}^\top \mathbf{X}$ is invertible.

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Takeaway

The inverse is a mathematical expression for the optimizer. It is not an instruction to form an inverse in numerical code.

Rank deficiency: different weights can give the same fit

When can full column rank fail? If $D > N$, or if feature columns are exactly linearly dependent.

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$$\mathbf{X}(\mathbf{w} + a(1, -1, 0, \dots, 0)^\top) = \mathbf{X}\mathbf{w} \quad \text{for every } a \in \mathbb{R}.$$

More generally, $\mathbf{X}\mathbf{v} = \mathbf{0}$ implies $\mathbf{X}(\mathbf{w}^* + \mathbf{v}) = \mathbf{X}\mathbf{w}^*$.

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Takeaway

Rank deficiency can yield multiple coefficient vectors with the same fitted values. The least-squares fitted vector is still the unique projection; the inverse formula is unavailable.

III-conditioning: how should we compute least squares?

The spectral condition number κ_2 measures numerical sensitivity: larger κ_2 means greater sensitivity, as with nearly dependent columns.

$$\kappa_2(\mathbf{X}^\top \mathbf{X}) = \kappa_2(\mathbf{X})^2 \quad \text{when } \mathbf{X} \text{ has full column rank.}$$

Forming $\mathbf{X}^\top \mathbf{X}$ squares the spectral condition number.

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Takeaway

The inverse formula explains the solution; numerical code should solve the least-squares problem with QR/SVD, for example `np.linalg.lstsq(X, y, rcond=None)`.

Do not explicitly form the inverse. A stable solver avoids unnecessary numerical error; it does not remove sensitivity in the data.

From mathematical optimum to computational choice

LIVE QA — pairs, 30 s: fact first, method second

Which fact about the problem should determine how we compute the fit?

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Very large or streaming data → mini-batch SGD: use cheap updates, then check convergence.

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Takeaway

Normal equations characterize the optimum; a factorization or iterative method computes it. Do not explicitly form $(\mathbf{X}^\top \mathbf{X})^{-1}$.

Session 1: explain the method and the residual

CHECKPOINT QA — 45 s solo, 60 s pair, then share

A design has two identical feature columns. A classmate proposes $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$. Explain (1) how to compute the fit, (2) whether weights are unique, and (3) where the residual points.

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Method choice: with very large data, mini-batch SGD may offer cheaper updates; a converged least-squares optimum satisfies the same normal equations. Next: why choose squared error in the first place?

Optional: arbitrary sampling

Session 2: What does the loss mean?

LIVE QA — think, vote, explain

We minimized squared error. What assumption about the observations makes that choice reasonable?

Gaussian noise: a model for repeated observations

Condition on the observed inputs $\mathbf{X}$. Assume

$$y_n = \mathbf{x}_n^\top \mathbf{w} + \epsilon_n, \quad \epsilon_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

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The noise is independent of the inputs; $\sigma^2 > 0$ is shared by all examples. The unknown weights determine the conditional mean:

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Takeaway

A prediction is the center of a distribution of possible observations. The noise model describes how observations vary around it.

One observation: how plausible is its residual?

Write $e_n = y_n - \mathbf{x}_n^\top \mathbf{w}$. The Gaussian density is

$$p(y_n \mid \mathbf{x}_n, \mathbf{w}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{e_n^2}{2\sigma^2}\right).$$

LIVE QA — vote σ or 2σ , then use the exponent

With the same σ , compare residuals $e_n = \sigma$ and $e_n = 2\sigma$. Which observation has the smaller density?

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We compare *densities*; a continuous variable has probability zero of taking any one exact value. Next, combine all N observations.

Probability model versus likelihood: what is varying?

Before observing y : probability model

$$p(y \mid \mathbf{x}, \mathbf{w})$$

Fix $(\mathbf{x}, \mathbf{w})$ and ask which responses y the model could generate.

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After observing $(\mathbf{x}, y)$: likelihood

$$L(\mathbf{w}; \mathbf{x}, y) = p(y \mid \mathbf{x}, \mathbf{w})$$

Fix the data and compare candidate weights $\mathbf{w}$.

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Let $y = 3$ and keep σ fixed. Candidate A predicts 2; candidate B predicts 0. Which candidate has larger likelihood?

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$|3 - 2| < |3 - 0|$, so $L(\mathbf{w}_A) > L(\mathbf{w}_B)$ under Gaussian noise. Likelihood *ranks* parameters; without a prior it is not a normalized probability distribution over parameters.

From one observation to the likelihood of the sample

With $\mathbf{X}$ fixed, conditional independence gives

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \prod_{n=1}^N p(y_n \mid \mathbf{x}_n, \mathbf{w}) = \prod_{n=1}^N \mathcal{N}(y_n \mid \mathbf{x}_n^\top \mathbf{w}, \sigma^2).$$

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Takeaway

Fix the observed data and vary $\mathbf{w}$. Maximum likelihood chooses weights that give these observations the largest joint density under the assumed model.

Take a logarithm: products become sums

The logarithm is increasing, so maximizing likelihood also maximizes log-likelihood. Negating it gives a minimization problem.

$$-\log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = -\sum_{n=1}^N \log p(y_n \mid \mathbf{x}_n, \mathbf{w})$$

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Takeaway

The negative log-likelihood is scaled squared error plus C .

Interpretation: Gaussian MLE recovers least squares

Recall our half-MSE $\mathcal{L}(\mathbf{w}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$.

$$-\log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \frac{N}{\sigma^2} \mathcal{L}(\mathbf{w}) + C.$$

LIVE QA — 15 s: vote yes/no, compare two candidates

Does adding C or multiplying by $N/\sigma^2 > 0$ change which weights minimize the objective?

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Squared error encodes a conditional Gaussian noise assumption. We can still use least squares without assuming that the model is true.

Transfer check: change the noise, derive the loss

Assume independent Laplace noise with fixed scale $b > 0$:

$$p(y_n \mid \mathbf{x}_n, \mathbf{w}) = \frac{1}{2b} \exp \left(-\frac{|y_n - \mathbf{x}_n^\top \mathbf{w}|}{b} \right).$$

CHECKPOINT QA — pairs, 45 s: log, simplify, name the loss

Take the negative logarithm of the joint likelihood. Which familiar loss has the same minimizers?

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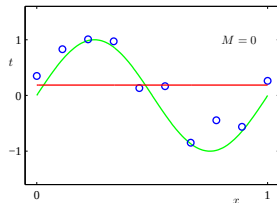
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Answer: mean absolute error (MAE). Noise $\rightarrow$ likelihood $\rightarrow$ negative logarithm $\rightarrow$ loss. Here b denotes the Laplace scale, not an intercept.

Can a good objective compensate for a limited model family?



Teaching simulation: $y_n = g(x_n) + \epsilon_n$.

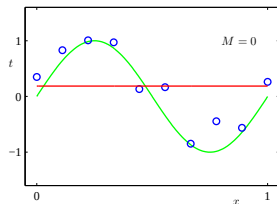
Blue circles: sampled observations.

Red curve: fitted model.

Green curve: generating function g .

The plot's vertical axis t denotes our response y .

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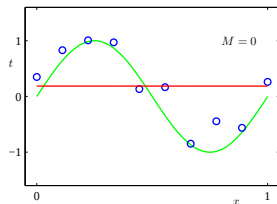
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The degree-zero model predicts a constant. Can changing that constant capture the rise and fall in the data?

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No. The family lacks flexibility. We know g only because this is a simulation; in practice we observe sampled data.

Mechanism: change the features, keep linear weights

For scalar input x , choose degree M and form a *column* vector:

$$\phi(x) = [1, x, x^2, \dots, x^M]^\top \in \mathbb{R}^{M+1}.$$

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$$f_{\mathbf{w}}(x) = \phi(x)^\top \mathbf{w} = w_0 + w_1x + \dots + w_Mx^M.$$

The function can curve in x while remaining linear in the weights. The constant feature supplies the intercept w_0 .

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The function can curve in x while remaining linear in the weights. The constant feature supplies the intercept w_0 .

Takeaway

Replace row $\mathbf{x}_n^\top$ of the design by $\phi(x_n)^\top$. We can use the same least-squares solvers to fit the new design.

What can increasing the degree guarantee?

Polynomial families are nested: $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots \subseteq \mathcal{F}_M$. A degree- M polynomial is also degree at most $M + 1$ by setting its new coefficient to zero.

CHECKPOINT QA — two votes: training error, then unseen error

Fit each family by globally minimizing the same training MSE. Can the minimum training error increase when we add a feature? What about error on unseen data?

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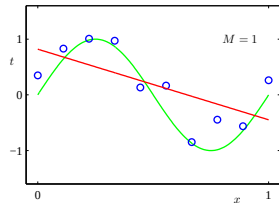
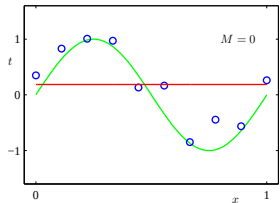
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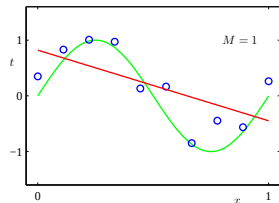
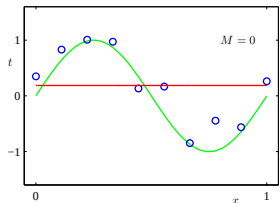
Takeaway

Lower training error alone gives no guarantee of better performance on unseen data. Flexibility can capture sample-specific variation.

Limited flexibility: inspect the training structure

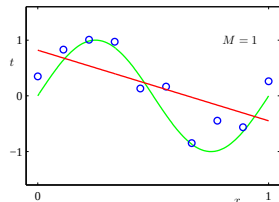
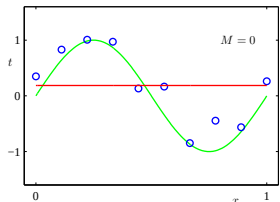


Limited flexibility: inspect the training structure



Observe: neither a constant ($M = 0$) nor a straight line ($M = 1$) captures the rise and fall of these observations.

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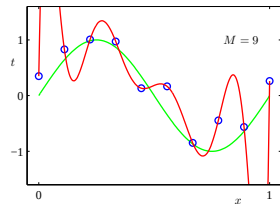
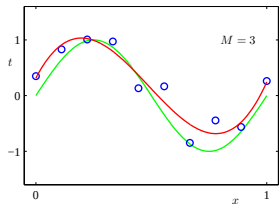


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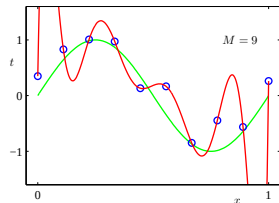
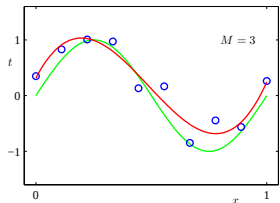
Takeaway

Under-fitting: an inflexible family leaves important training structure unexplained. We expect high error on similar unseen data as well; fitting the same family more accurately will not add curvature.

More flexibility: when the sample is fitted too closely

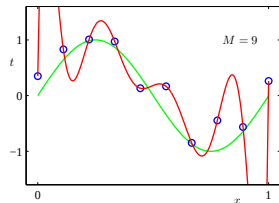
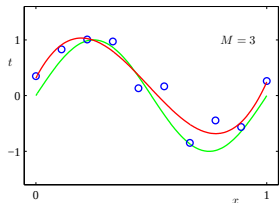


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Observe: $M = 3$ follows the broad pattern; $M = 9$ passes through all ten training points and oscillates sharply between them.

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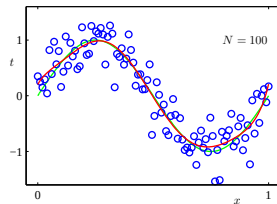
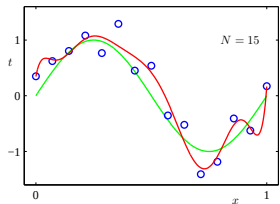


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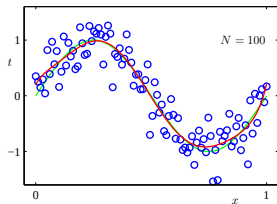
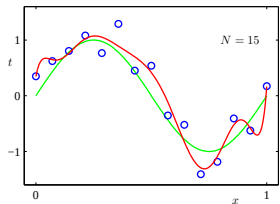
In this simulation, $M = 9$ fits sample-specific noise: its lower training error comes with worse expected unseen-data performance. Interpolation alone is not a universal proof of over-fitting.

Keep the degree fixed, change the amount of evidence



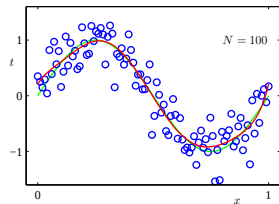
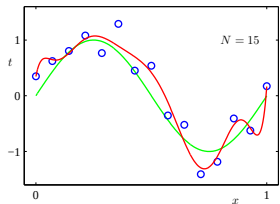
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Same degree $M = 9$: compare $N = 15$ and $N = 100$. More representative observations can constrain sample-specific fluctuations. The larger sample here supports a smoother fit.

This illustrates a possible benefit, not a guarantee for every sample. If more data are unavailable, how else could we control flexibility? Session 3 introduces regularization.

Define the problem through training and unseen data

LIVE QA — pairs, 30 s: write two diagnostic sentences

Use two pieces of evidence: what remains unexplained in training, and how sensitive is the fitted curve between observations?

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Takeaway

In real applications we have sampled data, not the generating curve. We need evidence beyond training error to diagnose generalization. Next lecture introduces validation, model selection, and bias-variance.

A diagnostic grid: what does the training sample tell us?

Inspect evidence before naming the behavior.

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Small training error alone does not establish unseen performance. These observations are diagnostic clues, not a measured test error.

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Takeaway

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Next action: try intermediate complexity or regularization; use validation in Lecture 04 to compare unseen-data estimates.

Session 2: diagnose two fits on a new noisy sample

CHECKPOINT QA — 45 s solo, 60 s pair, then share

On the same 20 observations, fit A has degree 1 and MSE 0.40; fit B has degree 19 and MSE 0.00. A leaves curved structure in its residuals; B oscillates between points. Compare flexibility, training fit, and expected unseen-data behavior. What would you try next?

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Decision: consider an intermediate degree or regularization.

Optional: MLE asymptotics

Session 3: How do we control complexity?

LIVE QA — think, vote, explain

A flexible model fits the training sample very well. How can we keep useful features while making its fitted weights less sensitive to this sample?

Regularization: keep the features, change the preference

Polynomial features gave us a richer model family. Large fitted weights can also make predictions sensitive to small changes in the data.

$$\min_{\mathbf{w}} \underbrace{\mathcal{L}(\mathbf{w})}_{\text{fit the observations}} + \underbrace{\lambda R(\mathbf{w})}_{\text{prefer smaller weights}}, \quad \lambda \geq 0.$$

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Takeaway

Regularization introduces bias and can reduce variance. Better unseen-data performance is a motivation, not a guarantee.

Before a penalty: set the units and separate the intercept

Use training-set means and standard deviations $s_j > 0$. For this session, $x_{nj} = x_{nj}^{\text{raw}}/s_j$ denotes a rescaled feature.

$$\mathbf{x}_{c,n} = \mathbf{x}_n - \bar{\mathbf{x}}, \quad y_{c,n} = y_n - \bar{y}, \quad \mathbf{X}_c = [\mathbf{x}_{c,1}, \dots, \mathbf{x}_{c,N}]^\top.$$

Thus each column of $\mathbf{X}_c$ is centered and standardized; y_c is centered. Remove constant columns before scaling.

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The intercept is not penalized. Reuse the training transformation for new inputs; b and $\mathbf{w}$ use the same rescaled feature units.

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Takeaway

Without scaling, coefficient penalties are not comparable: changing meters to centimeters changes the coefficient size.

Why units change a coefficient penalty

LIVE QA — vote yes/no; preserve prediction

The same length is recorded in meters or centimeters. Should that choice change how strongly we regularize its contribution?

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Takeaway

Standardization makes a common λ comparable across features. The intercept represents a baseline, so we keep it unpenalized.

L_2 -regularization: the ridge objective

On the centered, standardized design, penalize squared coefficient size:

$$J_\lambda(\mathbf{w}) = \frac{1}{2N} \|\mathbf{y}_c - \mathbf{X}_c \mathbf{w}\|_2^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2, \quad \|\mathbf{w}\|_2^2 = \sum_{j=1}^D w_j^2.$$

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What happens at $\lambda = 0$? What changes as the penalty strengthens?

At $\lambda = 0$ this is ordinary least squares (OLS). Increasing λ puts more weight on shrinking the coefficient vector; individual coordinates need not all change monotonically.

The $1/2$ in each term keeps the derivative simple. The intercept remains outside the penalty.

Explicit ridge solution: follow the factor N

Differentiating the same objective gives

$$\nabla J_\lambda(\mathbf{w}) = -\frac{1}{N} \mathbf{X}_c^\top (\mathbf{y}_c - \mathbf{X}_c \mathbf{w}) + \lambda \mathbf{w}.$$

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In code: use a linear-system solver (or augmented QR/SVD); the inverse above is mathematics, not an instruction to form it.

Ridge shrinks one eigen-direction at a time

Let $\mathbf{X}_c^\top \mathbf{X}_c \mathbf{v}_j = d_j \mathbf{v}_j$, with $d_j \geq 0$ and $\|\mathbf{v}_j\|_2 = 1$. Define $a_j = \mathbf{v}_j^\top \mathbf{X}_c^\top \mathbf{y}_c$.

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Takeaway

Small positive d_j means stronger shrinkage. If $d_j = 0$, then $a_j = (\mathbf{X}_c \mathbf{v}_j)^\top \mathbf{y}_c = 0$ and ridge sets that null component to zero.

Ridge lifts the small eigenvalues

Write the symmetric Gram matrix as $\mathbf{X}_c^\top \mathbf{X}_c = \mathbf{U} \text{diag}(d_1, \dots, d_D) \mathbf{U}^\top$, with $d_j \geq 0$ and orthogonal $\mathbf{U}$.

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Takeaway

The lift reduces unequal eigenvalue ratios and limits sensitivity along weakly supported directions. It also changes the fit.

Check: two nearly redundant sensors

CHECKPOINT QA — calculate 45 s; report one benefit and one cost

Two standardized sensor features are almost identical. Their Gram eigenvalues are 199.98 and 0.02. If $N = 100$ and $\lambda = 0.1$, predict the new condition number. Why might ridge help? What does it cost?

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The system is much less sensitive in the nearly redundant direction. **Decision:** ridge is a plausible choice if both sensors carry useful signal. Shrinkage trades some fit and bias for stability; this calculation alone does not establish better unseen predictions.

L_1 -regularization: the lasso

Suppose a small subset of the available features plausibly carries the signal.

$$\min_{\mathbf{w}} \quad \frac{1}{2N} \|\mathbf{y}_c - \mathbf{X}_c \mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_1, \quad \|\mathbf{w}\|_1 = \sum_{j=1}^D |w_j|.$$

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The data-fit term is still squared error. The change from ridge is the penalty: absolute values instead of squares.

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Takeaway

Lasso can produce a sparse, interpretable model. It does not guarantee exact recovery of the true relevant features. Features are standardized; the intercept is not penalized.

One-dimensional lasso: when does a coefficient become zero?

For one normalized coordinate, consider

$$q(w) = \frac{1}{2}(w - z)^2 + \lambda|w|, \quad \lambda \geq 0.$$

LIVE QA — sketch near zero, then predict

Think of z as the unregularized coefficient. What should happen when its magnitude is no larger than the penalty threshold λ ?

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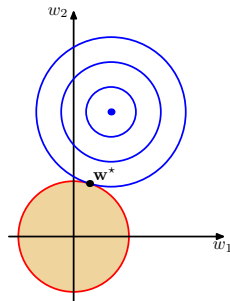
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This is exact for the stated one-dimensional (or orthonormal-coordinate) problem. Correlated multivariate designs do not share this one-shot formula.

Ridge geometry: a budget on squared coefficient size

Write $\mathcal{L}_c(\mathbf{w}) = \|\mathbf{y}_c - \mathbf{X}_c \mathbf{w}\|_2^2 / (2N)$.

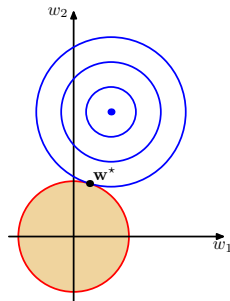


$$\min_{\mathbf{w}} \mathcal{L}_c(\mathbf{w}) \quad \text{subject to} \quad \|\mathbf{w}\|_2^2 \leq t.$$

Blue curves: equal unpenalized loss. The disk limits the weights.

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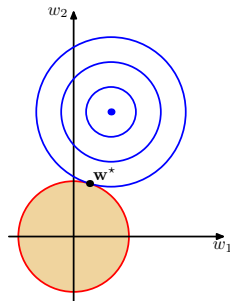
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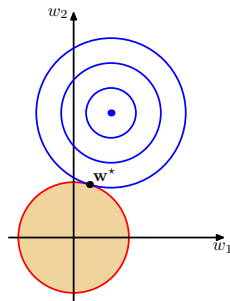
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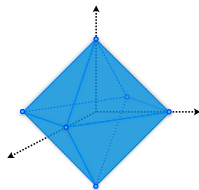
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Each penalized solution has a corresponding budget $t \geq 0$. Here t bounds squared norm; the disk radius is $\sqrt{t}$.

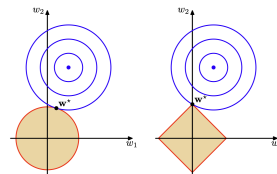
t and λ are not the same numerical parameter. Stronger penalties generally mean tighter budgets.

Lasso geometry: corners lie on coordinate axes

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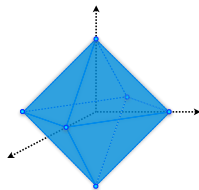
A three-dimensional L_1 ball.



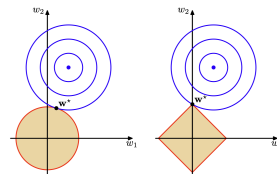
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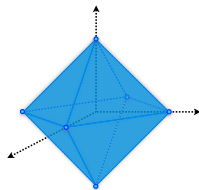
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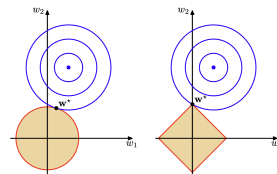
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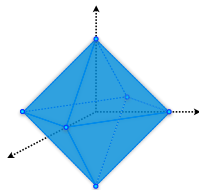
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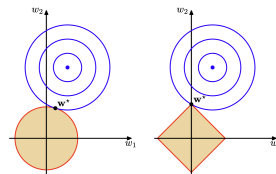
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Exact zeros are common, not guaranteed for every dataset and penalty. The budget t corresponds to a penalized solution; it is not λ .



Smooth L_2 boundary versus L_1 corners.

Check: does a zero coefficient identify the truth?

CHECKPOINT QA — vote yes/no, then reason about correlation

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Takeaway

Prefer lasso when sparsity is a defensible modeling assumption. For many correlated predictors with dense signal, ridge is often a more natural starting point.

From likelihood to MAP: add a belief about weights

Start with $y_n = b + \mathbf{x}_n^\top \mathbf{w} + \epsilon_n$, conditional on the inputs:

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}), \quad \sigma^2, \tau^2 > 0.$$

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evaluated at the optimal intercept.

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Prior: prefer smaller weights

$$-\log p(\mathbf{w}) = \frac{\|\mathbf{w}\|_2^2}{2\tau^2} + C'$$

Smaller τ^2 expresses a stronger preference for zero.

MAP becomes ridge with a specific penalty strength

Bayes' rule: $p(b, \mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{X}, b, \mathbf{w}) p(\mathbf{w})$ with a flat intercept prior; optimize over b as on the previous frame.

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Divide by $N/\sigma^2 > 0$: the minimizer is unchanged.

$$\frac{1}{2N} \|\mathbf{y}_c - \mathbf{X}_c \mathbf{w}\|_2^2 + \frac{\sigma^2}{2N\tau^2} \|\mathbf{w}\|_2^2 = J_\lambda(\mathbf{w}), \quad \boxed{\lambda = \frac{\sigma^2}{N\tau^2}}.$$

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Takeaway

Gaussian noise plus a Gaussian weight prior gives ridge MAP. For fixed N , more noise or a tighter prior means stronger shrinkage. No penalty is applied to the intercept.

Session 3: choose a method for a new problem

CHECKPOINT QA — 60 s solo, 90 s pair; method + assumption

Choose OLS, ridge, or lasso for each case. Give a reason tied to conditioning, shrinkage/generalization, or sparsity.

- 1 Six well-conditioned predictors; ample data and no reason to shrink.
- 2 Many correlated predictors; small effects spread across most of them.
- 3 Hundreds of candidate predictors; domain knowledge supports a small subset.

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A complete answer: name the method and justify its assumption. Standardize features before penalties; keep the intercept unpenalized. Select λ and compare unseen performance through validation in Lecture 04 (next lecture), alongside generalization and bias–variance.

Optional: Rayleigh proof

Deepening: a Gaussian vector

OPTIONAL

For $\mathbf{y} \in \mathbb{R}^N$, the ordinary multivariate Gaussian density requires a *positive definite* covariance Σ :

$$p(\mathbf{y}) = \frac{\exp \left[-\frac{1}{2} (\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{y} - \boldsymbol{\mu}) \right]}{(2\pi)^{N/2} \det(\Sigma)^{1/2}}.$$

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In our conditional model, $\boldsymbol{\mu} = \mathbf{X}\mathbf{w}$ and $\Sigma = \sigma^2 \mathbf{I}$ with $\sigma^2 > 0$. The density then factors into the N scalar densities used in Session 2.

A singular positive semidefinite covariance defines a degenerate Gaussian, but the inverse/determinant density above does not apply.

[Return to Session 2 check](#)

Optional: MLE in large samples

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For iid observations, the *average* log-likelihood estimates an expectation:

$$\frac{1}{N} \sum_{n=1}^N \log p(y_n \mid \mathbf{x}_n, \mathbf{w}) \longrightarrow \mathbb{E}_{x,y}[\log p(y \mid x, \mathbf{w})].$$

Suppose the model is correctly specified and identifiable; $\mathbf{w}_{\text{true}}$ is an interior point; a uniform law of large numbers and regular, integrable derivatives apply; and the information below is nonsingular. Then a consistent MLE obeys

$$\mathbf{w}_{\text{MLE}} \xrightarrow{p} \mathbf{w}_{\text{true}}, \quad \sqrt{N}(\mathbf{w}_{\text{MLE}} - \mathbf{w}_{\text{true}}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{I}_1(\mathbf{w}_{\text{true}})^{-1}).$$

[Return to Session 2 check](#)

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For iid observations, the *average* log-likelihood estimates an expectation:

$$\frac{1}{N} \sum_{n=1}^N \log p(y_n \mid \mathbf{x}_n, \mathbf{w}) \longrightarrow \mathbb{E}_{x,y}[\log p(y \mid x, \mathbf{w})].$$

Suppose the model is correctly specified and identifiable; $\mathbf{w}_{\text{true}}$ is an interior point; a uniform law of large numbers and regular, integrable derivatives apply; and the information below is nonsingular. Then a consistent MLE obeys

$$\mathbf{w}_{\text{MLE}} \xrightarrow{p} \mathbf{w}_{\text{true}}, \quad \sqrt{N}(\mathbf{w}_{\text{MLE}} - \mathbf{w}_{\text{true}}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{I}_1(\mathbf{w}_{\text{true}})^{-1}).$$

$\mathbf{I}_1(\mathbf{w}) = \mathbb{E}[-\nabla_{\mathbf{w}}^2 \log p(y \mid x, \mathbf{w})]$ is per-observation Fisher information. Under the same regularity, it equals the score covariance. Asymptotically, covariance is approximately $\mathbf{I}_1^{-1}/N$; efficiency is among regular estimators and is not a finite-sample equality.

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Deepening: two other ways to see convexity

OPTIONAL

Composition argument. Each $\mathcal{L}_n(\mathbf{w}) = \frac{1}{2}(y_n - \mathbf{x}_n^\top \mathbf{w})^2$ is a convex square composed with an affine function. A nonnegative average of convex functions remains convex.

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Deepening: two other ways to see convexity

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Definition argument. For $\lambda \in [0, 1]$,

$$\begin{aligned} \mathcal{L}(\lambda \mathbf{w} + (1 - \lambda) \mathbf{w}') - \lambda \mathcal{L}(\mathbf{w}) - (1 - \lambda) \mathcal{L}(\mathbf{w}') \\ = -\frac{\lambda(1 - \lambda)}{2N} \|\mathbf{X}(\mathbf{w} - \mathbf{w}')\|_2^2 \leq 0. \end{aligned}$$

Expand the squares to verify the identity. Equality along a nonzero direction is possible if that direction is in the null space of $\mathbf{X}$.

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Optional: arbitrary-sampling SGD

OPTIONAL

Let $f(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N f_i(\mathbf{w})$. Sample I with probabilities $p_i > 0$ (not necessarily uniform). The importance-weighted estimate at the *current* iterate is

$$\hat{\mathbf{g}}_I(\mathbf{w}^{(t)}) = \frac{1}{Np_I} \nabla f_I(\mathbf{w}^{(t)}).$$

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For m iid draws, average these m weighted estimates. The update is $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \hat{\mathbf{g}}(\mathbf{w}^{(t)})$.

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For m iid draws, average these m weighted estimates. The update is $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \hat{\mathbf{g}}(\mathbf{w}^{(t)})$. Uniform sampling has $p_i = 1/N$, recovering ∇f_I . Non-uniform sampling needs the $1/(Np_i)$ factor to stay unbiased.

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Optional: Rayleigh proof of ridge invertibility

OPTIONAL

For a symmetric matrix $\mathbf{A}$ and $\mathbf{v} \neq \mathbf{0}$, define

$$R(\mathbf{A}, \mathbf{v}) = \frac{\mathbf{v}^\top \mathbf{A} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}}, \quad \lambda_{\min}(\mathbf{A}) = \min_{\mathbf{v} \neq \mathbf{0}} R(\mathbf{A}, \mathbf{v}).$$

With $\mathbf{A} = \mathbf{X}^\top \mathbf{X} + N\lambda \mathbf{I}$ and $\lambda > 0$,

$$R(\mathbf{A}, \mathbf{v}) = \frac{\|\mathbf{X}\mathbf{v}\|_2^2 + N\lambda\|\mathbf{v}\|_2^2}{\|\mathbf{v}\|_2^2} \geq N\lambda.$$

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Therefore $\lambda_{\min}(\mathbf{A}) \geq N\lambda > 0$: the ridge system is positive definite and invertible, even when $\mathbf{X}$ is rank-deficient. This is the Rayleigh-quotient version of eigenvalue lifting.

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